



Fig. 2: REM-free technologies such as AEM's HDRM can be scaled to cover all vehicles from three-wheelers to busses, without compromise to performance (Courtesy: AEM, Sterling GTake)

windings. AEM's high-density reluctance motor (HDRM) and super-speed reluctance drive (SSRD) offer performance comparable to traditional permanent magnet motors, while also addressing issues such as cost volatility, recyclability, and environmental impact.

Dr James Widmer, co-founder of AEM, emphasised during a recent virtual session titled 'Rare Earth Magnet-Free Motors: Next Generation Electric Traction' in India, that the EV industry still largely follows legacy designs from the 1990s. "We are offering something better and more sustainable," he said. AEM's motors are already powering commercial vehicles in Europe and have logged over 4 million kilometres. Their partnership with Sterling is now focused on localising production in India, tapping into the growing interest in magnet-free solutions.

In contrast to other Indian innovations, AEM's approach is more radical. It eliminates both magnets and copper, replacing copper with

aluminium. While TVS, Ola Electric, and Mahindra Electric are working on varying levels of localisation and rare-earth reduction, AEM's method represents a complete technological shift rather than an incremental improvement.

James detailed the reasons driving this shift. "Rare-earths are expensive and their prices are volatile due to geopolitical tensions. With 100 million EVs projected annually by the 2040s, it is neither economically nor environmentally sustainable to depend on such materials." Rare-earth mining often involves radioactive elements like thorium, and James warned that, if EV produc-

tion scales without change, rare-earth mining could generate more low-grade nuclear waste than the nuclear power industry.

He further highlighted the challenges related to recyclability and supply chain fragility. "Recovering rare-earths and copper from end-of-life motors is extremely difficult," he noted. In response, AEM has developed motors using recyclable materials. Their patented aluminium coil technology allows for lightweight, efficient, and scalable production. The HDRM300, initially designed for buses and trucks, delivers performance equivalent to permanent magnet motors, without using magnets or copper, resulting in a 60% weight reduction.

AEM has also miniaturised the HDRM for compact applications. "A German Tier 1 supplier commissioned us to develop a smaller version, again, rare-earth-free, half the size of the nearest competitor for the same output," said James. The SSRD platform is currently under development for both the Indian and

European passenger car markets. He emphasised that HDRM maintains high efficiency across all speeds, unlike traditional motors, and cited a customer report indicating a 12% improvement in vehicle range.

Addressing concerns around noise, James clarified that any difference is negligible. The use of aluminium in place of copper was also deliberate. Although aluminium has lower conductivity, its use significantly reduces weight and improves overall efficiency, due to AEM's patented coil design.

Jaideep Wadhwa, Director of Sterling Tools Limited and Managing Director of Sterling GTake E-Mobility Limited, echoed James's concerns during the same session. He questioned the viability of investing heavily in rare-earth mining in India. "China has a 45-year head start.

Developing that capacity domestically would take a decade, and it is a very polluting industry," he said, quoting a *New York Times* report, "2000 tonnes of waste are produced for every tonne of rare-earths mined."

Instead, Wadhwa advocated for India to leapfrog this legacy model. "This is 30-year-old technology. If you look at what we were driving three decades ago, there is little of that you would want to drive today," he said. He acknowledged that AEM's motors may not be suitable for every application, but insisted they offer a robust and scalable alternative. "There are enough alternative technologies that allow us to avoid rare-earth dependence. The challenge is to develop and nurture them."

On the broader issue of electrification, Wadhwa was unequivocal, "Yes, we must pursue it. Our crude oil import bill is \$80 billion and continues to rise. We cannot make oil, but we can build electronics." He also confirmed that one motor platform has already been localised